

SCRIBING | LECTURE 08

Classification

K-nearest neighbours, hierarchical clustering, and supervised learning

Course	BMEG 3105 — Data analytics for personalized genomics and precision medicine
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Outline of lecture

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 - A. Similarity and dissimilarity
 - B. Hierarchical clustering and linkage rules
- II. Classification
 - A. Why classification? What is it?
 - B. Training data, a classification method, and a query
- III. K-nearest neighbour (KNN) classification
 - A. Workflow and design choices
 - B. Worked example: classify P5 with $K = 2$
- IV. Clustering versus classification
- V. Self-study and next topics

Lecture in one sentence

Clustering discovers groups in unlabelled data; classification uses labelled examples to assign a known class to a new record.

I. Review and bridge from clustering

A. Similarity and dissimilarity

Clustering groups records so that within-cluster differences are small and between-cluster differences are large. A similarity or distance measure gives the algorithm a way to compare records. The lecture lists cosine similarity, correlation, Euclidean distance, Manhattan distance, and Mahalanobis distance; Mahalanobis distance accounts for the distribution and covariance of the data.

B. Hierarchical clustering

Hierarchical clustering builds nested clusters and represents their sequence of merges as a dendrogram. Such trees can describe meaningful taxonomies, including gene clusters, phylogenies, or biological kingdoms.

Standard procedure

1. Compute a similarity or distance matrix.
2. Start with each data point as its own cluster.
3. Merge the two closest clusters.
4. Update the matrix using a chosen linkage rule, then repeat.
5. Stop when all points have merged into one cluster; inspect the dendrogram or cut it to obtain groups.

Common ways to update inter-cluster distance

Linkage rule	How distance between two clusters is defined
Minimum (single)	Smallest pairwise distance across the two clusters
Maximum (complete)	Largest pairwise distance across the two clusters
Group average	Average of pairwise distances across the two clusters
Centroid	Distance between the clusters' centroids

C. Gene-expression example

The lecture compares five genes across wild type and mutant conditions using linear correlation. Each gene begins as a singleton cluster. The first merge is At2g34930 with At1g30720 (correlation 0.9909). The worked sequence next adds At4g35770, then At2g05540, with At4g27450 joining last. The slides update the matrix by selecting the strongest correlation (equivalent to the smallest dissimilarity under that convention).

Important distinction

When a matrix stores distances, smaller is closer. When it stores similarities or correlations, larger is closer. Keep the convention consistent when choosing each merge.

II. Classification

A. Motivation and definition

Classification is useful for organizing items and deciding where a new item belongs. In biology, a new gene-expression profile might need to be classified as Normal or Tumor. Other examples include grouping patients for different treatment considerations or identifying customers in a target group.

Given a training set of records with attributes and a known class label, classification learns a method that assigns a class to previously unseen records using their other attributes as accurately as possible.

II. Classification (continued)

B. What is needed?

1. Training data with class labels.
2. A classification method learned or selected from the training data.
3. A new, unlabelled record to classify.

Example: use labelled records P1–P4 (height, weight, and gender) to predict the unknown gender of P5 from height and weight. The target class is withheld from the model input and supplied as the prediction output.

III. K-nearest neighbour classification

A. Core idea and workflow

KNN classifies a query by comparing it with stored labelled instances. Its intuition is: “If it walks like a duck and quacks like a duck, then it is probably a duck.” Training mainly stores the examples; prediction finds the K closest examples and returns the most frequent class among them (the mode).

1. Normalize numeric features so different units or scales do not dominate the distance.
2. Compute each training point’s distance from the query using the chosen metric.
3. Select the K nearest training points.
4. Return the mode class among those neighbours (optionally weight votes by distance).

Choices to make: a distance metric (e.g., Euclidean, Manhattan, Mahalanobis, correlation or cosine similarity), the number of neighbours K, and optionally a weighting function. The lecture notes that K around 5–10 can work well for many low-dimensional datasets; cross-validation is a more principled way to choose K.

B. Worked example: classify P5

Assume Euclidean distance and $K = 2$. The lecture’s table min–max normalizes height and weight separately: $x' = (x - \text{minimum}) / (\text{maximum} - \text{minimum})$.

Person	Height (m)	Weight (kg)	Gender
P1	1.79	75	M
P2	1.64	54	F
P3	1.70	63	M
P4	1.88	78	M
P5 (query)	1.75	70	?

After min–max normalization, the coordinates and distances from P5 are:

Person	Height'	Weight'	Gender
P1	0.6250	0.8750	M
P2	0.0000	0.0000	F
P3	0.2500	0.3750	M

Person	Height'	Weight'	Gender
P4	1.0000	1.0000	M
P5 (query)	0.4583	0.6667	?

III. K-nearest neighbour (continued)

C. Distance calculation and prediction

For each training person, Euclidean distance is computed in the two-dimensional normalized feature space: $d(P_i, P_5) = \sqrt{[(\text{height}'_i - \text{height}'_5)^2 + (\text{weight}'_i - \text{weight}'_5)^2]}$.

Person	Distance to P5	Class
P1	0.267	M
P2	0.809	F
P3	0.358	M
P4	0.636	M
P5 (query)	0.000	Unknown

The two closest labelled records are P1 (0.267, M) and P3 (0.358, M). Their mode class is M, so KNN predicts M for P5.

Result

With Euclidean distance, min–max normalized features, and $K = 2$, P5 is classified as M.

IV. Clustering versus classification

Aspect	Clustering	Classification
Goal	Discover similarity-based groups in data	Assign a known class to new data
Input	Unlabelled data	Labelled training data + unlabelled test/query data
Classes	Number of groups is generally unknown	Set of possible classes is known
Output	Cluster index for each record	Predicted class for each new record
Process	One-stage grouping procedure	Training phase followed by application/prediction

V. Supervised and unsupervised learning

Unsupervised learning analyzes unlabelled data to find structure; clustering and dimensionality reduction are examples. Supervised learning trains on labelled examples to predict outcomes; classification and regression are examples.

Key takeaways

- Classification learns from labelled examples; clustering discovers groups without labels.
- KNN is instance-based: normalize, measure distance, select neighbours, and vote.
- The distance/similarity convention and the choice of K affect the result.

Self-study and next topics

Suggested reading: Introduction to Data Mining, Chapters 4.1, 4.2, and 5.2. Topics identified in the lecture for further study include KNN limitations, logistic regression, neural networks, decision trees, SVMs, Bayesian methods, overfitting, and cross-validation.

Source note: Prepared from the supplied Lecture 8 “Classification” slides by Yu Li. The provided scribing PDF was used as a structural and stylistic reference; its separate logistic-regression content was not substituted for this lecture’s classification material.